The material appearing on this website is for informational purposes only. It provides the material author's perspective on a wide variety of economic and financial system issues.

FEDS Notes

April 03, 2017

Robustness of long-maturity term premium estimates

Calvin L. Ballard and Marius Rodriguez

Introduction

In order to shed light on the reasons for movements in Treasury yields, we often decomposes yields into the expected path of the federal funds rate and an additional term premium, which compensates investors for holding bonds. In order to do this, we need a model that accurately accounts for the dynamics of the term market prices and we need to estimate it using econometric models. As highlighted in recent commentary by the FT,[1] there is a degree of uncertainty about both the best model to use for these purposes and the robustness of the resulting term premium estimates.

This Note explores the difference between two of the models used to produce estimates of US Treasury term premiums which are produced by staff in the Federal Reserve System. The first is a model developed by Adrian, Crump, and Moench (2013) which is currently used by the Federal Reserve Board and the FOMC meeting.[2] The other is the model used by staff at the Federal Reserve Bank of New York by Adrian, Crump and Mabrouk (2013) (ACM). The structures of these models are very similar but Figure 1 shows that their long maturity term premium estimates have considerably different volatility at times. Analysis in this Note suggests that the main reason for the divergence between model estimates is that the Adrian, Crump, and Moench model (ACM) exhibits considerably more volatility than is justified by the data used to estimate the model, once we modify the ACM model to include three surveys, term premium estimates are very similar to the ACM estimates.

Figure 1. Three-Month and Adrian Crump Moench (acm) 10-year term premiums

Accessible version

One advantage of including surveys is that this can address some of the small sample problems associated with estimating term premiums using only yields. Since the dynamics of the term market is hard to estimate the level to which short-term interest rates will eventually revert in the speed of mean reversion from yields alone, which can result in substantial estimation uncertainty and – as noted by the FT – a degree of uncertainty about the robustness of the estimates. Adding data from the Survey of Primary Dealers can provide the model with more information about short-term interest rate expectations, thereby increasing the precision of term premium estimates and making them less sensitive to the choice of sample period.

That is not to say that the more precise estimates from models that include surveys are necessarily more accurate. Rather, it is possible that the survey data contains a different term premium than the marginal bond participants. If that is the case, one might prefer an impressive estimate from a model that omits survey or a precise but biased estimate. Ultimately, however, further work to evaluate a wider range of models and to better understand the survey data will be required to better understand the term premium.

Given the substantial model and (in some cases) parameter uncertainty involved with estimating dynamic term structure models, it is worth bearing in mind that while some features of estimated term premiums appear to be fairly robust across models (e.g. long maturity term premium, extrapolations from the models considered in this Note are all top histories statistics), other features (such as the sign and volatility of short-term premiums) are not.

Why do estimates of term premia differ?

The KW and ACM models published by the Federal Reserve System have much in common. They both belong to the class of so-called Gaussian no-arbitrage affine term structure models (Duffie and Kan 1996; Dai and Singleton 2000). In these models, bond yields are driven by a small number of pricing factors. In both the KW and ACM models, these factors are assumed to follow a Gaussian vector-autoregressive process (VAR) and to jointly determine the short-term interest rate. The short-term rate implies a unique path (which we can think of as a proxy for the federal funds rate) is modeled as a linear function of the pricing factors. Given the VAR and the short-rate process, the prices of all bonds in the model are then set to determine the relationship between the pricing factors and yields. In addition, the model allows for a wedge between the long-run yield that is equal the average of the five year yield and the five-year, five-year forward rate or there would be an arbitrage opportunity.

Given estimates of the model parameters, we can compute term premiums as the difference between the observed yield and the model-implied yield. In both the KW and ACM models, the model-implied yield is the expected future short-interest rate over the relevant bond maturity, which is easily computed by formulating the VAR of pricing factors.[3] In these models, interest rate distributions are Normal (and therefore symmetric) and the expectation of future short-term interest rates is equal to the median outcome. The shadow rate approach proposed by Black (1995) are likely to be more appropriate for analyzing term premiums at short maturities during recent years. In this FEDS Note we therefore focus on term premiums at the 10-year maturity which should be less affected by the lower bound on nominal rates.[4]

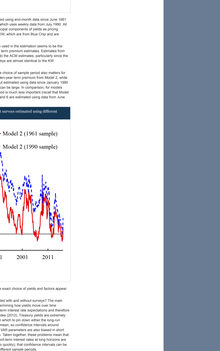
Table 1 at the end of this Note summarizes the differences between KW and ACM. First, the parameters governing the VAR are estimated using different data sets. The KW model uses a term structure literature) uses maximum likelihood to estimate all the parameters of the model in one go. While maximum likelihood is statistically efficient, it is numerically very challenging because of the large number of parameters and the highly nonlinear nature of the model's likelihood function. Alternatively, the ACM splits up the estimation into multiple linear regressions, which is computationally much faster.[5]

Second, KW and ACM use "pricing factors yields differently: they use yields of different maturities, have different number of pricing factors, and use different measurement assumptions. The KW model uses 3 pricing factors, weekly data, and makes different choices of sample period (ACM uses a sample starting in June 1961, KW in July 1990). Third, ACM is estimated using yields data on Treasury yields while KW also uses data on Blue Chip survey expectations of future short-term interest rates in addition to yields (note that the KW model is not constrained to match the survey expectations exactly; rather, it allows for them to be informative about the term premium).

To investigate which of these differences contributed to the diverging term premium estimates in Figure 1, we estimate three variations of ACM that are estimated using only monthly data on Treasury yields (shown in Figure 2):

1. the ACM estimates from a free factor model published by the NY Fed;
2. a three-factor model estimated using maximum likelihood (i.e. a KW-style regression method); and
3. a three-factor model estimated using maximum likelihood.

Figure 2. Ten-year term premium estimates from models without surveys

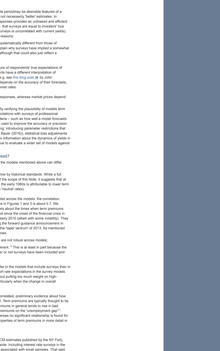


Accessible version

as well as three variations of KW that also incorporate surveys in the data set (shown in Figure 3):

4. the three-factor KW model estimates using weekly data published by the Board;
5. a three-factor model using monthly data estimated using an extension of the ACM method that allows up to incorporate surveys;[6] and
6. a three-factor model estimated using maximum likelihood (i.e. very similar to KW) but using monthly data.

Figure 3. Ten-year term premium estimates from models with surveys



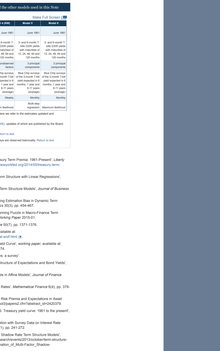
Accessible version

Table 1 provides further details on the models. All are estimated using end-month data since June 1961 with the exception of Model 4 (the published KW estimates), which uses weekly data from July 1990. All models include a 3-month term premium and extrapolate to a 10-year term premium using the "Kim-Wright factors." Models 5 and 6 include the same set of surveys as KW, which are from Blue Chip and are incorporated in the estimation from 1983.

We have three main findings. first, whether or not surveys are used in the estimation seems to be the most important factor in explaining the differences in 10-year term premium estimates. Estimates from models that include surveys are far less volatile and much lower than the ACM estimates, particularly since the mid-1990s, while estimates from the models that include surveys are almost identical to the KW estimates.

Second, particularly for models estimated without surveys, the choice of sample period also matters for term premium estimates. The last row of Table 1 shows the ten-year term premium from Model 2, while the blue, dotted line shows estimates from the same model but estimated using data since January 1990 rather than June 1961 and then extrapolated backwards using the parameters and pricing factors estimated using surveys. Figure 3 shows that the sample period is not very important for the Model 6 is estimated using data from July 1990, whereas Models 5 and 6 are estimated using data from June 1961.

Figure 4. Ten-year term premium from models without survey estimated using different sample periods



Accessible version

Finally, other factors such as the estimation technique and the exact choice of yields and factors appear to be much less important.

How do we explain these differences between models estimated with and without surveys? The main reason is that it is challenging to estimate the parameters determining how yields move over time without data on short-term interest rate expectations. Although the dynamics of the term structure of interest rates are fairly well-understood, as discussed by Kim and Orphanides (2005), Treasury yields are extremely persistent, so there is little information in a typical sample with which to pin down either the long-run mean of the short-term interest rate or the speed at which short-term interest rates are expected to revert. Small sample estimates of term premiums can be volatile.[7] Over estimates of VAR parameters are also biased in short samples and typically over-state the speed of mean reversion. Taken together, these problems mean that models estimated without survey data may attribute movements in Treasury yields to changes in the term premium when they are in fact due to short-rate expectations. Survey data on short-term interest rate expectations can help to pin down the long-run mean and the speed of mean reversion.

Although survey measures of expectations are not frequently updated (e.g. the long-term survey used in the KW model is only published biannually), including them in the data set for estimating term structure models can be very important for estimating the dynamics of the term premium. Survey data can help pin down short rate and how quickly short-term interest rates should revert back to their long-run mean (Kim and Wright 2005; Kim and Orphanides (2012). As a result, term premium estimates from survey models are less sensitive to the choice of sample period than models without survey data. Although the survey expectations compared here sometimes and term premiums beta substantially, we find that they typically help to offset some of the small sample bias mentioned above (less variable long-horizon term premiums are like flat-slope of nominal variable yield curve).

The sensitivity to the sample period in models that do not include surveys is problematic in part because estimates of the term premium using the entire sample suggest that it was relatively low and uncommon in the term structure literature. A sample that starts shortly after the 1980s reduces the likelihood that structural breaks associated with changes in monetary policy will bias parameter estimates and – as noted by Kim and Orphanides (2012) – long-maturity forward rates may be a biased proxy for the almost no information in yields with which to estimate the long-run average and the speed of mean reversion. Using a longer sample means that there is potentially more information to estimate the parameters but it also increases the probability that the parameters have changed. For example, the FT commentary mentioned above suggested it is "implausible" that after 1970s inflation expectations rose, expected yields to fall over time as inflation was rising. Mechanically, this is simply a consequence of the fact that the 10-year yield was lower than the 3-month yield for much of the 1980s and implies that investors were expecting them to fall back towards the mean. The sensitivity of the models that include survey only do as from 1983 onwards but it is worth noting that short-maturity surveys from 1983 suggest that short-term interest rates were indeed expected to fall over the subsequent year.

Are surveys a panacea?

While precise estimates that are robust to changing the sample performing the desirable features of a model, it is important to remember that precise estimates are not necessarily "better" estimates. In particular, if the survey data used to estimate the term structure models does not accurately reflect measures of expectations of Treasury market participants (i.e. that surveys are equal to investors' true expectations on average and that the measurement error is uncorrelated with current yields), then estimates of term premiums from models that include surveys will be biased.

- Expectations of Blue Chip survey respondents may be systematically different from those of Treasury-market traders. It is possible that this could explain why surveys have implied a somewhat faster pace of lift-off than market rates in recent years (although that could also reflect a negative term premium at short maturities).
- It is possible that survey forecasts are not a good measure of respondents' true expectations of future short-term interest rates, or that survey respondents have a different interpretation of "short-term interest rates" than the overnight federal funds rate (Cole and Reis 2017; Coibion/Gorodnichenko). For example, if respondents' rationalization depends on the persistence of their forecasts, surveys may contain a risk premium just like market interest rates.
- Surveys are computed as equal-weighted averages of responses, whereas market prices depend on the views of the marginal investor.

Ultimately, it is challenging to find convincing ways of externally verifying the plausibility of models term premiums. One approach is to compare short-rate expectations with surveys of professional forecasters provides one reasonable approach – but other criteria – such as how well a model forecasts are also relevant. There are also other methods that could be used to improve the accuracy or precision of term premium estimates without relying on surveys, including: introducing parameter restrictions that are based on economic theory (Ang, Dong, and Piazzesi 2007); using more flexible specifications of shocks (Kaus, Rudebusch and Wu (2012); or incorporating non-price information about the dynamics of yields in a Bayesian setting. In future work, we therefore plan to continue to evaluate a wider set of models against a wider set of criteria than discussed in this Note.

What features of term premiums appear to be robust?

Despite the substantial differences between estimates from the models mentioned above can offer, some features do appear to be quite robust across models:

- Estimates of long-maturity term premiums are currently low by historical standards. While a full discussion of the potential explanatory factors is beyond the scope of this Note, it is important that at least part of the decline in long-term interest rates since the early 1980s is attributable to lower term premiums (rather than just being about lower short-term "neutral" rates).
- Changes in estimated term premiums are highly correlated across the models. The correlation between monthly changes in the 10-year term premium is at least 0.7 in all cases. It is also therefore just a broadly consistent story across the models about the times when term premiums were falling or rising. For example, focusing on the period since the onset of the financial crisis in 2007, all models suggest that term premiums fell during the financial crisis, rose during the taper "tantrum" (May 2013 to end-2013) and again following the forward guidance announcement in August 2011, before rising for a period during and after the taper tantrum of 2013. As mentioned above, they have since fallen back to historically low levels.

Or in other words, the broad story about the cyclical behavior of term premiums is robust.

- The sign of term premiums at short maturities is sometimes in dispute. This is at least in part because the average level of the term premium is sensitive to whether or not surveys have been included and the choice of sample period.
- The magnitude of movements in term premiums is smaller in the models that include surveys than in the models that do not. We find the flip side of more variable short-rate expectations in the survey models is that the term premium is less variable. Incremental caution about putting too much weight on high-frequency changes in estimates of term premiums – particularly when the change in overall Treasury yields is relatively small.
- Although the term premium estimates are quite highly correlated, particular models can differ significantly in the size of the term premium cycle. For example, Model 1 "the premiums are typically thought to be counterintuitive because the magnitude of asset risk premiums in general tends to rise in bad times." Simple (unreported) regression of ACM term premiums on the "unemployment gap" suggests that the term premium is countercyclical in the sample used to estimate the model. Similar for the KW estimates. We plan to investigate the cyclical properties of term premiums in more detail in future work.

Conclusion

In models that are estimated using yields only (such as the ACM estimates published by the NY Fed), parameters governing the dynamics of short-term interest rates are difficult to estimate precisely. Adding the estimation of term structure models can reduce the problem associated with small samples. That said, while robustness to changing the sample period and matching surveys reasonably closely are desirable features of a model, they are not the only criteria by which we should evaluate a model. In future work, we evaluate a wider set of models against a wider set of criteria would be helpful for moving forward the debate. At least for the time being, however, it is always worth considering the robustness of signals from a range of different approaches when making statements about term premiums.

Table 1. Summary of differences between ACM, KW and the other models used in this Note

	Model 1 (ACM)	Model 2	Model 3	Model 4 (KW)	Model 5	Model 6
Sample	June 1961 – present	June 1961 – present	June 1961 – present	July 1990 – present	June 1961 – present	June 1961 – present
Estimation method	Free factor model, 3 pricing factors, 18 parameters, 18 or 20 lag VAR, 18 parameters, 18 or 20 lag VAR	Free factor model, 3 pricing factors, 18 parameters, 18 or 20 lag VAR	Free factor model, 3 pricing factors, 18 parameters, 18 or 20 lag VAR	3 pricing factors, 18 parameters, 18 or 20 lag VAR	3 pricing factors, 18 parameters, 18 or 20 lag VAR	3 pricing factors, 18 parameters, 18 or 20 lag VAR
Pricing factors	5 principal components	3 principal components	3 principal components	3 principal components	3 principal components	3 principal components
Survey?	No	No	No	Blue Chip survey of 10-year 3-month T-bill yields, 3-month T-bill yields, 1-year and 2-year T-note yields	Blue Chip survey of 10-year 3-month T-bill yields, 3-month T-bill yields, 1-year and 2-year T-note yields	Blue Chip survey of 10-year 3-month T-bill yields, 3-month T-bill yields, 1-year and 2-year T-note yields
Frequency	Monthly	Monthly	Monthly	Weekly	Monthly	Monthly
Estimation method	Multi-step OLS/Kalman filter	Multi-step OLS/Kalman filter	Maximum likelihood	Maximum likelihood	Multi-step OLS/Kalman filter	Maximum likelihood
Sample	1 is the published sample, ACM uses a sample starting in January 1961					

1 is the published sample, ACM uses a sample starting in January 1961 there are also updated and published on the NY Fed website and Bloomberg. Return to text

2 Survey forecasts are taken from the Survey of Primary Dealers, updates of which are published by the Board. Return to text

3 Principal components of yields with maturities 1, 6, ..., 120 months. Return to text

4 Kim and Wright (2005) uses an alternative monthly, long-term surveys are observed biannually. Return to text

References

Adrian, T., R. A. Crump, E. Mills and E. Mainch (2014), "Treasury Term Premia: 1961-Present", *Liberty Street Economics* (blog), http://libertystreeteconomics.newyorkfed.org/2014/05/treasury-term-premia-1961-present.html.

Adrian, T., R. A. Crump and E. Moench (2013), Pricing the Term Structure with Linear Regressions", *Journal of Financial Economics* 110(1), pp. 110-138.

Bauer, M. D. (2016), "Restrictions on Risk Prices in Dynamic Term Structure Models", *Journal of Business and Economic Statistics*, forthcoming.

Bauer, M. D., G. D. Rudebusch and J. C. Wu (2012), Correcting Estimation Bias in Dynamic Term Structure Models", *Journal of Business and Economic Statistics* 30(3), pp. 444-457.

Bauer, M. D. and G. D. Rudebusch (2016) "Resolving the Spanning Puzzle in Macro-Finance Term Structure Models", *Federal Reserve Bank of San Francisco Working Paper* 2016-01.

Black, F. (1995) "Interest Rates as Options", *Journal of Finance* 50(1), pp. 1371-1376.

Cochrane, J. H. (2014) "Real or not-neutral rate", blog post available at http://johnhcochrane.blogspot.com/2014/ Cochrane-or-not-neutral-rate.html.

Cochrane, J. H. and M. Piazzesi (2008), "Decomposing the Yield Curve", working paper, available at http://faculty.chicagobooth.edu/john.cochrane/research/Papers/.

Council of Economic Advisers (2015), "Long-term Interest Rates", January.

Crump, R. K., Eusepi, S. and Moench, E. (2016), "The Term Structure of Expectations and Bond Yields", *Federal Reserve Bank of New York Staff Reports* 775.

Duffie, G. R. (2002), "Term Premia and Interest Rate Forecasts in Affine Models", *Journal of Finance* 57(1), pp. 405-443.

Duffie, D. and R. Kan (1996), "A Yield-Factor Model of Interest Rates", *Mathematical Finance* 6(4), pp. 379-406.

Gürkaynak, R. (2016) "Fed's Emest? Robust Identification of Fed Premia and Expectations in Asset Prices", *Bilkent University and Federal Reserve Board*, presentation available at http://www.bilkent.edu.tr/~refik/Fed%20Emest.pdf.

Gürkaynak, R. S., B. P. Sack and J. H. Wright (2007) "The U.S. Treasury yield curve: 1961 to the present", *Journal of Monetary Economics* 54(2), pp. 2291-2304.

Kim, D. H. and A. Orphanides (2012), "Term Structure Estimation with Survey Data on Interest Rate Forecasts", *Journal of Financial and Quantitative Analysis* 47(1), pp. 241-272.

Kim, D. H. and J. H. Wright (2005), "An Arbitrage-Free Three-Factor Term Structure Model and the Recent Behavior of Long-Maturity Forward Rates", *Finance and Economics Discussion Series 2005-33. Board of Governors of the Federal Reserve System*, August.

Kim, D. H. (2016) "What's up with the Term premium?", *Financial Times*, 1 December 2016.

Mark, S. and A. Mamaev (2016) "Evaluating the robustness of UK term structure decompositions using linear regression methods", *Journal of Banking and Finance* 67, pp. 85-102.

Sam, A. and A. Apostolov (2016), "Zero-lower bound and the 10-year term premium", *UK Global Rates Strategy Focus*, 28 November 2016.

[1] Mark (2016), who draws on the results of Sam and Apostolov (2016). Return to text

[2] See Council of Economic Advisers (2015) and Kim (2016). Return to text

[3] https www.bloomberg.com/news/articles/2016-11-01/interest-rate-and-accelerating-markets

https://www.bloomberg.com/news/articles/2016-11-01/interest-rate-and-accelerating-markets

[4] Although the shadow rate approach is likely to be more appropriate, the ACM model is known to be estimated using Normal filtering techniques. This does not make a material difference to estimates of term premiums in level. Return to text

[5] ACM actually uses a slightly different definition of the term premium. But this differs from the definition we use only by a small Jensen's inequality adjustment and is quantitatively very similar. Return to text

[6] A full discussion of the pros and cons of different estimation techniques is beyond the scope of this Note. We focus on attention only on the implications for estimates of long-horizon term premiums. Return to text

[7] In ACM, the VAR governing the time-series dynamics of the pricing factors is estimated by OLS. Yields are highly persistent, so the VAR coefficients are very close to unity. Although OLS is unbiased, it is not efficient and the estimated coefficients are biased toward the mean. Taking the resulting short-rate dynamics as fact, we then estimate the term premium by pinning down the long-run average yield such that it equals the average five-year yield. Because the measurement of expected future short-rate dynamics depends on the estimated coefficients, small changes in coefficients rather than one month period can be enough to assume that the data have a material difference. The intercepts of the ACM estimation are very sensitive to the sample period.

[8] This result is consistent with the Bauer and Rudebusch (2016) finding that the Jensen's inequality term in the term premium is driven by the average of the ACM term premium and is measures compared by simply subtracting survey expectations of future short-term interest rates. Bauer and Rudebusch (2016) also suggest the projections based on survey-based term premium estimates is more robust. Return to text

[9] Of course all of this assumes that it WAS with constant parameters is the most appropriate model for the dynamics of yields. It is possible that the dynamics of the term structure have changed over time. For example, the Federal Reserve's asset purchase devices a long-duration short-term "neutral" interest rate expectations in recent years. Return to text

[10] A positive term premium suggests that investors view bonds as a risky asset – i.e. they tend to have relatively low pay-offs in bad times (when marginal utility is high). Cochrane (2014) and Cochrane and Piazzesi (2008) find that bonds do not hedge against bad times (bonds pay-offs are bad in good times). Return to text

[11] The argument depends on the term premium being positive, which is tautological, as discussed above. Return to text

Please cite this note as:
L. Calvin, Andrew Meldrum, and Marius Rodriguez (2017), "Robustness of long-maturity term premium estimates," FEDS Notes. Washington: Board of Governors of the Federal Reserve System, April 3, 2017. https://doi.org/10.17016/2380-7172.1927.

Disclaimer: FEDS Notes are articles in which Board staff offer their own views and present analysis on a range of topics in economics and finance. These articles do not reflect the views of the Federal Reserve System or of the Federal Reserve System's staff members.

Last Update: February 21, 2018

USA.gov GOTENBERG